AP CALCULUS AB

Unit 1 • Lesson 1.5 notes

Lesson 1.5 • Determining Limits Using Algebraic Properties of Limits

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

Learning goals

- Use the sum, difference, constant multiple, product, quotient and power properties of limits.
- Evaluate limits by direct substitution when the function is continuous at the point.

Vocabulary. limit properties, direct substitution, composite function.

1. Properties of limits

Key idea

If $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$ (real numbers), then

- $\lim(f \pm g) = L \pm M$, $\quad \lim(kf) = kL$, $\quad \lim(fg) = LM$;
- $\lim \dfrac{f}{g} = \dfrac{L}{M}$ if $M \neq 0$; $\quad \lim f^n = L^n$; $\quad \lim \sqrt[n]{f} = \sqrt[n]{L}$ (with $L \geq 0$ for even n).

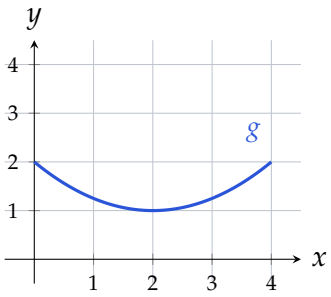
2. Direct substitution

Polynomials, rational functions (where the denominator is not 0), roots, sin, cos, e^x and $\ln x$ (on their domains) satisfy $\lim_{x\to c} f(x) = f(c)$.

Example 1

Solution. $\lim\limits_{x\to -1} \left(x^3 - 4x + 2\right) = -1 + 4 + 2 = 5$; $\quad \lim\limits_{x\to \pi/3} (2\cos x) = 2 \cdot \frac{1}{2} = 1$; $\quad \lim\limits_{x\to 1} \dfrac{e^x}{x+1} = \dfrac{e}{2}$.

Example 2 • Using given limits and a graph



$\lim_{x\to 2} f(x) = -3$, and the graph of g is shown. Find $\lim_{x\to 2} \left(f(x)\, g(x) - x^2\right)$.

Solution. From the graph, $\lim_{x\to 2} g(x) = 1$. So the limit is $(-3)(1) - 4 = -7$.

Common mistakes

- Applying the quotient property when the denominator's limit is 0.
- Using the product property when one of the limits does not exist.

Quick check

$\lim_{x\to 5} f(x) = 2, \lim_{x\to 5} g(x) = -4$. Find: 1. $\lim(3f - g)$ 2. $\lim \frac{g}{f^2}$ 3. $\lim \sqrt{f+7}$

Answers. 1. 10. 2. -1. 3. 3.